Forecasting HDB Rental Rates in Singapore

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***Abstract*—Rent estimates provide a benchmark for assessing asking rents in Singapore's public housing market. We predict monthly rents for 50,000 Housing and Development Board (HDB) rental approvals from March 2025 to July 2026 using 150,000 approvals from January 2021 to March 2025. Because 99.2% of test records occur after the last labeled month, prediction requires estimating future market conditions as well as differences between flats. We estimate a multiplicative monthly market index to distinguish common rent movements from changes in the block and flat-type composition of rental approvals. We forecast the index with a damped trend and train LightGBM and CatBoost to predict rent divided by the index. Their predictions are multiplied by the forecast index and averaged with histogram-based gradient boosting (HistGB) trained on unadjusted rent. Removing index adjustment increases rolling-validation root mean squared error (RMSE) by S$86.4 in a matched LightGBM comparison. Across eleven model classes, RMSE spans S$32.2 under a fixed index setup. Rows sharing all recorded predictors with another rental constitute 27.6% of validation observations. On this subset, estimated pooled within-group variance equals 98.8% of one LightGBM configuration's mean squared error. This estimate does not establish an error floor for other rentals or the hidden test set. The ensemble achieves a best public-leaderboard RMSE of S$430.76.**

***Index Terms*—rental price prediction, market index forecasting, gradient boosting, temporal validation, within-group variation**

I. Introduction

HDB rents rose steeply after 2021, raising housing costs for households that do not own a flat, including those not yet eligible to buy one and those waiting for a new flat to be completed [1]. A rent prediction lets a prospective tenant compare an asking rent with the expected rent of flats with similar recorded attributes. Landlords and real estate agents can use rent estimates to set competitive asking prices. The Singaporean government can use adjusted rent trends to inform housing supply and rental assistance policies.

The task is to predict the monthly rent of 50,000 Housing and Development Board (HDB) rentals approved between March 2025 and July 2026 from 150,000 labeled rentals approved between January 2021 and March 2025. The test set is separated from the training set in time rather than sampled at random, and 99.2% of its rows fall after the last labeled month. Over the labeled period the mean monthly rent rose from S$2,049 in January 2021 to S$3,163 in March 2025, an increase of 54%. A model must therefore predict the market level up to 16 months beyond the training data as well as the differences between flats within a month.

The temporal and cross-sectional components require different treatment. A standard regression tree partitions the training feature range into regions with constant predictions. With other predictors held fixed, values beyond the largest observed month follow the same branches, so adding a month feature cannot extrapolate a continuing trend. Other inputs, including time-dependent ages and macroeconomic variables, can still change predictions. Cross-sectional accuracy is limited by the recorded attributes. Rentals of the same flat type in the same block and month have identical predictors but different rents. Section VIII estimates the variation within these groups.

We investigate how to predict future market levels while estimating rent differences associated with recorded flat attributes. We compare location representations and model classes to assess which information and modeling choices improve prediction. We also examine variation among rentals with identical predictors to characterize limitations of the available attributes. Figure 1 summarizes the method. Rent is modeled as a base component determined by recorded flat attributes multiplied by a monthly market index. Chronological validation tests both components over the forecast horizon.

II. Data, Exploratory Analysis, and Preprocessing

A. Task and Data Sources

The task is a Kaggle competition [1] based on HDB rental approval records published on data.gov.sg [2]. Each row records the approval month, town, block, street, flat type, flat model, floor area, lease commencement year, furnishing status, and upfront fee. Training rows also record the monthly rent in Singapore dollars. Submissions are evaluated by RMSE. Rent-error metrics are reported in Singapore dollars unless expressed as ratios.

The records have a grouped panel structure, with block–flat-type groups recurring across months. Individual flats cannot be identified, so records sharing these attributes are not assumed to represent the same dwelling. Temporal features summarize group history rather than repeated observations of an identified flat.

The competition supplies six auxiliary tables. They contain block characteristics and coordinates for 9,464 blocks and the locations of 243 Mass Rapid Transit (MRT) station records, 113 shopping malls, and 337 schools. They also contain monthly Certificate of Entitlement (COE) premiums for four vehicle categories and daily share prices of 77 Singapore-listed companies. Amenity locations provide geographic accessibility proxies, block characteristics describe building composition, and economic series are evaluated as predictors of market movements. Only the economic series vary with approval month and extend into the test period.

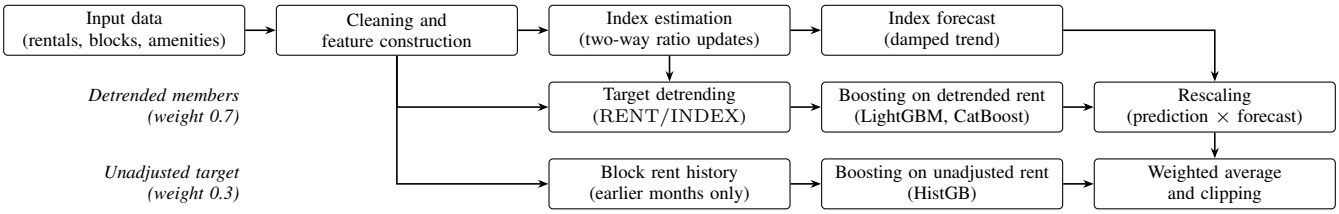


Figure 1. The pipeline combines two index-adjusted models with a model trained on unadjusted rent. The top row constructs features, estimates monthly market indices by flat type, and forecasts them over the prediction window. LightGBM and CatBoost learn rent divided by the fitted index, and their predictions are multiplied by the forecast index. HistGB learns unadjusted rent using earlier block–flat-type rents as an additional feature. The ensemble assigns weight 0.7 to the mean index-adjusted prediction and 0.3 to HistGB. All components are refitted using each outer fold's training data.

We added the HDB Property Information register [3], which covers 13,357 blocks and records dwelling counts by flat type, rental-unit counts, and block facilities. It contains no transaction rents, so the join does not supply target values. As a static register, it may not represent each block's historical condition. The public rental approval register was excluded because its coverage extends to July 2026 and includes test-period rents.

B. Cleaning and Target Distribution

Neither the training file nor the test file contains missing values. Two columns are constant. Column `FEE` equals zero and column `FURNISHED` equals "yes" in all 200,000 training and test rows. A constant column cannot distinguish one rental from another, so both are dropped. Column `FLAT_TYPE` has eleven distinct values in the training file because each numbered type is written in two formats, for example "4 room" and "4-room", whereas the test file uses only the hyphenated format. After every value is mapped to the hyphenated format, six flat types remain. The 1-room type is rare, with 42 training rows.

Rent ranges from S$300 to S$7,600, with a mean of S$2,719 and a standard deviation of S$750. The median rent increases from S$1,450 for 1-room flats to S$3,000 for 5-room and executive flats, but floor area alone explains only about 11% of the variance. The annual mean rent was S$2,121 in 2021, S$3,036 in 2023, and S$3,110 in 2024, so most of the increase occurred before 2024.

We examined whether extreme rents indicated errors and whether logarithmic transformation improved prediction. A Tukey fence, defined by extending 1.5 interquartile ranges below the first and above the third quartile, flags 752 rents (0.5%) outside S$450–S$4,850. One third have other records with the same flat type, block, and month. The median absolute deviation from their group mean, which includes the flagged rent, is S$333. This comparison does not establish that the records are erroneous, so they were retained. Rent has skewness $+0.39$, whereas log rent has skewness -0.42. The log-target configuration increases RMSE by 5.4 and the magnitude of mean error from S$18.0 to S$67.9, so subsequent models use rent in levels. Figure 3 shows both distributions and the concentration of rents at rounded amounts.

Joining the block table on town, block, and street initially matched only 79.5% of training rows, and all mismatches had the same cause. The block table stores street names in lower case, whereas 20.5% of rows in the rental files use title case, for example "Ang Mo Kio Avenue 3" and "ang mo kio avenue 3". Converting street names to lower case raises the match rate to 100%.

The inconsistent case also affected every quantity grouped by street and block, because the two spellings were treated as different blocks. The training file contains 17,413 distinct combinations of town, block, and street but only 9,406 physical blocks, and 8,007 of these blocks appear under both spellings. Without normalization, the rental history of most blocks would be split in two, which would distort both the block history feature of Section VI-B and the index units of Section V.

C. Variation by Month and Location

We first assessed variation associated with approval month and location. Replacing each rent by its approval-month mean explains 32.1% of sample variance. Using town–flat-type means explains 21.9%. Combining the groupings explains 57.6%. These descriptive, in-sample calculations indicate complementary temporal and cross-sectional information. They motivate separate modeling of the market level and flat attributes, but do not establish independence or constant relative prices.

Variation also remains among records with identical predictors. In this dataset, floor area, flat model, and lease commencement year are constant within each block–flat-type group. Rentals with the same block, flat type, and month therefore agree on every competition predictor. A deterministic model using these predictors must assign them the same prediction, making the group mean the minimum-squared-error fit to the observed group. A repeated feature group contains at least two records with identical predictors. Such groups contain 26.8% of training rows. Their observed within-group RMSE is S$343. Correcting for estimation of one mean per group gives a pooled noise standard deviation of S$464, approximately 17% of mean rent. The estimator and assumptions are given in Appendix H. This variation can reflect unobserved attributes or other noise [5]. For example, seven 3-room, 65-square-metre rentals in block 4 Holland Close approved in January 2025 have rents of S$3,300, 3,200, 3,050, 3,000, 2,800, 2,200, and 1,000.

III. Validation Protocol

We use chronological validation to reflect the prediction of future rental approvals.

Training months precede validation months, reflecting prediction of future approvals rather than interpolation within the observed period [6], [7]. Splits occur at month boundaries so that records from the same block and month cannot appear on both sides. Each rolling validation window covers the 16-month future forecast horizon. Quantities derived from rent, including the index, block history, and target encodings, are recomputed from each fold's training rows. This prevents validation rents from entering fitted features or index forecasts.

We use rolling-origin evaluation [6]. The training period expands to March, July, and November 2023, and each fitted model predicts the following 16 months (Table 4). Each validation fold contains approximately 47,000 rows. We report mean fold RMSE and mean prediction error, defined as prediction minus observed rent, to distinguish overall error from systematic over- or under-prediction. A fourth origin after March 2024 provides 12 labeled months. Candidates were retained only when they improved both the rolling mean and this confirmation cut. The confirmation cut therefore contributes to model selection and is not an independent test.

The validation windows overlap, and the confirmation window lies within the third rolling window. Their scores are consequently dependent. Difficulty also varies by period. One configuration scores approximately 555, 519, and 483 across the rolling folds. Models are compared on identical observations within each fold. Scores obtained under different windows are not directly comparable.

IV. Baselines and the Extrapolation Problem

We first established two baselines. The first is ordinary least squares (OLS) regression on floor area, lease age in years, town, and flat type. We use one-hot encoding for town and flat type, representing each category with a binary indicator that equals one when a record belongs to that category and zero otherwise.

A linear model with additive terms assigns each town a single premium that is added to the rent of every flat type, so a town that adds S$300 to a 3-room flat must also add S$300 to an executive flat. It cannot represent an interaction effect, in which the town premium depends on flat size. Hedonic price theory models a dwelling's price as a function of structural and location attributes without requiring additivity [8]. We therefore also fit histogram-based gradient boosting (HistGB) [9], [13]. Each tree fits residual errors from the preceding ensemble, with a learning rate scaling its correction. A split on town followed by a split on floor area allows the town premium to vary with size. Histogram binning reduces numeric split candidates. HistGB also handles categorical columns natively and assigns missing values to a learned branch.

The boosting baseline uses 12 features. These comprise the four OLS attributes, flat model, three amenity distances, two economic series, a sequential month counter, and calendar month. Distances from each block to the nearest MRT station, mall, and primary school are calculated from coordinates using the haversine formula [14]. Economic features are monthly mean COE premiums and share prices. Street and block are excluded because their high cardinality and median of 13 training rows per block create a risk of overfitting.

OLS uses months after the 85th percentile of approval dates. HistGB uses the latest 15% of date-sorted rows. Their RMSEs are 700.33 and 464.79, respectively. These initial holdouts differ and are shorter than the forecast horizon, so their scores are excluded from the main rolling-fold comparisons.

Permutation importance [15] measures the decrease in the coefficient of determination (R^2) after shuffling one feature on held-out data. Town and flat type have importances of 0.352 and 0.261, followed by floor area at 0.134 and nearest-MRT distance at 0.012 (Table 6). Correlated predictors complicate interpretation of individual importance scores [16]. These values measure reliance of the fitted model rather than causal contributions to rent.

The month counter has zero importance. The two economic series have marginally negative importance. Each learned threshold on the month counter lies within its training range, and every held-out value is at or beyond that range. Permuting these values therefore leaves the corresponding branches unchanged. This result concerns splits on the month feature. Predictions may still vary with other inputs. It motivates an explicit forecast of the market level rather than reliance on the month counter alone.

V. Multiplicative Market Index

A. Decomposition of Rent

For rental record i approved in month t, we use the approximation

$$\mathrm{RENT}_{it} \approx \mathrm{BASE}_i \, \mathrm{INDEX}_t. \tag{1}$$

The dimensionless index represents the common monthly rent level. LightGBM and CatBoost learn rent divided by the fitted index, and their predictions are multiplied by its forecast. For example, a rent of S$2,000 observed at an index of 0.8 gives an adjusted target of S$2,500. A predicted base of S$2,500 combined with a forecast index of 1.1 gives a rent prediction of S$2,750. Adjustment reduces market-level changes without establishing stationarity of the full target distribution.

Increasing dispersion motivates proportional rescaling. Estimated within-group noise standard deviation rises from S$279 in 2021 to S$588 in 2023. An additive alternative subtracts a dollar-valued month effect, estimated by alternating group and month mean-residual updates, and adds its forecast after regression. Both alternatives use a global monthly component, identical structural and block-composition features, fixed LightGBM settings, and the same slope window and damping. Target-based features are excluded. Rolling RMSE is 519.41 for multiplicative adjustment and 525.56 for additive adjustment. Confirmation RMSEs are 467.04 and 470.68. The settings were selected for the multiplicative pipeline, so this comparison does not independently tune the additive alternative.

To adjust for observed dwelling characteristics [8], let $g(i)$ identify the street–block–flat-type group of approval i, and let t_i be its month. Starting with $I_t = 1$, twelve iterations alternate the arithmetic means of ratios

$$B_g \leftarrow \operatorname*{mean}_{i:g(i)=g} \frac{y_i}{I_{t_i}}, \qquad I_t \leftarrow \operatorname*{mean}_{i:t_i=t} \frac{y_i}{B_{g(i)}}. \tag{2}$$

Here y_i is observed rent, and B_g is the group base used for index estimation. Each iteration of Equation (2) normalizes the last training-month index to one. This adjustment prevents a changing mix of groups from being attributed entirely to market movements. It assumes approximately stable group bases and does not remove unobserved changes within groups. Alternative grouping definitions change RMSE by at most 0.75.

In matched ablations, training on unadjusted rent increases rolling RMSE by 86.4 for LightGBM and 140.0 for CatBoost. Mean errors are S$−207 and S$−333, respectively. These results support explicit adjustment for the market level under the evaluated forecast windows.

B. Forecasting the Index

The index in Equation (1) must be forecast up to 16 months beyond the final training month. A random-walk forecast holds its last value. A linear forecast extends the least-squares slope b fitted over the last w months. The damped forecast [18] is $\widehat{I}_{T+h} = I_T + b\sum_{j=1}^{h}\phi^j$, where T is the last training month, h is the forecast horizon, and $0 < \phi < 1$ reduces successive increments. We crossed w and ϕ on the rolling folds to separate their effects.

The damping parameter has the largest effect. With the slope fitted on twelve months, ϕ values of 0.6, 0.75, 0.85, and 0.95 give RMSEs of 519.87, 521.41, 530.04, and 567.73, while the mean error increases from -4 to $+190$. Weak damping projects the earlier rise into a period in which rents had stopped rising. An undamped linear trend over 18 months gives 672.55, the worst index-forecast configuration tested. The slope window has a smaller effect, since the best setting of each window lies within 0.6 of the overall best. We therefore fit the slope on twelve months with ϕ of 0.6.

On the full training index, the selected forecast increases by 0.3% over 16 months, compared with 2.7% for the undamped 18-month trend (Figure 2). In the tuned window–damping comparison, the selected forecast reduces RMSE from 523.68 for a random walk to 519.87. By contrast, the 86.4-point ablation compares index adjustment with training LightGBM directly on rent. The distinction separates the effect of detrending from the choice of index forecast.

This conclusion depends on the validation period. All three validation windows begin after the 2022 rise, so each fold favours a forecast that expects rents to flatten, and a renewed steep rise would penalize a ϕ of 0.6. Across the forecasts considered, from a random walk to an undamped linear trend, the mean index over the test rows varies by 1.5%, or S$40 of rent (Table 9). This range does not bound the effect on the public score, because the level shift grows with the horizon and its relation to the errors on unlabeled rows is unknown.

A separate ridge regression, an L2-regularized linear model, uses 71 features and unadjusted rent and scores 505.60 publicly. Both its model and temporal specification differ, so comparison with our 430.76 ensemble does not isolate the effect of index adjustment.

C. Segment-Specific Indices

A global index assumes proportional market movements across flat types. Separate fitted indices increase from January 2021 to March 2025 by factors ranging from 1.46 for 2-room flats to 1.61 for executive flats. This difference motivates flat-type-specific indices.

Separate segment indices allow different growth rates but can increase estimation variance in small groups. For example, 2-room flats contribute only 3,098 rentals. We therefore shrink each segment index toward the global index geometrically, giving segment g the index $\mathrm{INDEX}_t(\mathrm{INDEX}_t^g/\mathrm{INDEX}_t)^\lambda$, so that $\lambda = 0$ recovers the global index and $\lambda = 1$ the fully separate one. Segmenting by flat type with $\lambda = 0.5$ gives 519.27 against 519.87 for the global index. Regional segmentation gives 519.47, a smaller improvement.

The final configuration gives RMSEs of 518.94, 518.82, 518.80, and 518.81 for $\lambda = 0.3, 0.5, 0.7, 0.9$, respectively. We selected 0.5 because validation accuracy is insensitive to this range. The 0.7 version scores 430.76 on the public leaderboard, compared with 431.01 for the reported 0.5 version. The observed difference is 0.25 RMSE. Its statistical uncertainty cannot be estimated from the submission files alone because the hidden labels, and hence the paired prediction errors, are unavailable.

VI. Feature Engineering

A. Amenities and Block Composition

We test whether accessibility and building characteristics add predictive information beyond flat attributes and administrative location. Eleven feature groups describe amenity distances and counts, coordinates and city-centre distance, block height and age, remaining lease, floor area and room count, administrative areas, and calendar month. Each group is removed in turn from the tuned model to assess its contribution.

Amenity features use supplied locations rather than a dated operational network. The MRT table has 243 rows at 192 locations, including repeated interchange coordinates and a record marked open for the planned station serving Changi Airport Terminal 5 [4]. Deduplicating and removing that station leaves 191 locations and changes RMSE by 0.02. This check does not reconstruct the network at each approval date, so MRT features are geographic proxies with limited interpretation as access to operating stations.

Removing any single point-of-interest group changes RMSE by at most 0.17 from the 519.87 reference. Removing coordinates improves it by 0.23. Removing all ten point-of-interest columns increases RMSE by 1.58, and removing them together with coordinates increases it by 2.24. These results indicate overlapping predictive information. A one-at-a-time

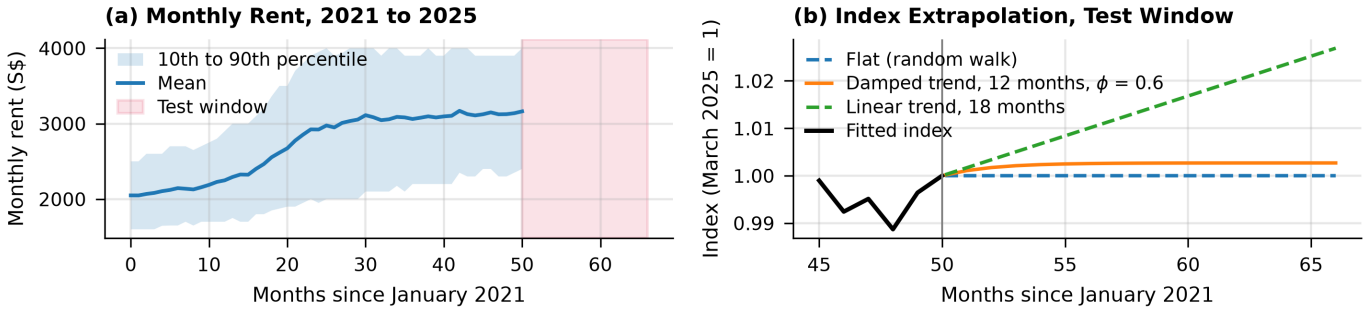


Figure 2. Panel (a) shows monthly mean rent and the band between the 10th and 90th percentiles during the labeled period. The shaded region marks the test period. Panel (b) compares the last five months of the fitted index with three forecasts, normalized to one in March 2025.

ablation understates their joint contribution. Twenty additional distance and count columns improve RMSE by 0.26. The largest individual removal effects are 5.29 for calendar month and 0.64 for floor area.

Seven features from HDB Property Information describe block composition. They are the dwelling count, the rental, small-unit, and large-unit shares, and indicators for commercial units, a hawker centre, and a multistorey car park. They reduce rolling RMSE from 519.87 to 519.27 and confirmation RMSE from 467.06 to 466.41. They supplement location with information about block composition. The register is unmatched for 159 training rows in 11 blocks and no test rows. These feature values remain missing and are handled by the boosting models.

A horizon feature, the number of months between the last training month and the predicted row, increased the RMSE by 14.3 and was rejected. Within the training data this quantity is confounded with calendar time, so the model used it as a proxy for the date.

Coordinate clustering tests whether regions defined by proximity improve on administrative areas. Replacing these areas with k-means clusters gives RMSEs of 519.46, 519.47, and 519.24 for $k = 8, 25, 60$, versus 519.87. Cluster centers are fitted within each training fold. Principal component analysis (PCA) tests whether correlated numeric predictors can be compressed without losing accuracy. It forms orthogonal linear combinations ordered by explained variance after median imputation and standardization. These transformations are fitted on training rows. Eight and sixteen components increase RMSE by 2.90 and 0.46, while twenty-four match the original 25 variables. The largest cluster-based reduction is 0.63 RMSE.

B. Block Rent History

We added two features to HistGB trained on unadjusted rent. These are the mean earlier rent for the same street, block, and flat type, and the number of contributing observations. These describe a block–flat-type group rather than an identified individual flat. Of the test rows, 99.2% have such history in the training data, with a median count of 15.

Both features are computed from the target, so they cause target leakage unless each value uses only rents that would be known at prediction time. A training row therefore receives the expanding mean over strictly earlier months, excluding its own month, so it uses neither its own rent nor the rent of a flat approved in the same month. A validation or test row receives the mean over all training rows, which is the information available at prediction time, when the most recent label is up to 16 months old. Rows without history keep a missing value, which the boosting model assigns to a learned branch, and the count indicates how reliable the mean is.

Block history lowers the RMSE by 12.1 on the rolling folds and by 13.7 on the confirmation cut, and it improves every fold. The improvement comes entirely from rows with history, since the two models perform equally on the 1 to 3% of fold rows without one. This model scored 436.05 on the public leaderboard, and Table 7 gives the results by fold.

Block history reduces mean under-prediction from S$80 to S$49. Earlier rents contain information about both the block premium and historical market levels. The reduction in bias is consistent with history partially compensating for the absence of explicit detrending, but does not isolate the source of the RMSE gain.

C. Target Encoding of Location

After detrending, target encoding tests whether historical location premiums add information beyond coordinates. It replaces categories with smoothed target means [17] at seven groupings involving town, street, block, subzone, and flat type. Without recency weighting, a group with n observations and adjusted-rent mean $\bar{r}$ receives $(n\bar{r} + 12m)/(n + 12)$, where m is its broader parent mean. Three observations therefore give 20% weight to the group mean and 80% to its parent. Training group aggregates exclude the row's approval month, and validation encodings use only outer-training rents. Spatial priors are observation-weighted adjusted-rent means from nearby blocks. The search retrieves 11 and 31 blocks and excludes the target block when present.

The configuration without hierarchical target encoding scores 523.32, compared with 529.30 when it is included. Changing the smoothing weight or using recency weights with 12- or 24-month half-lives changes RMSE by at most 1.8. All remain worse than the reference. A half-life is the interval after which an observation's weight is halved.

We tested whether the gap reflects the age of available history. Each training row received block history truncated a random 1–16 months before its approval month, approximating forecast-time staleness. RMSE increased by a further 12.5. This intervention did not improve accuracy and does not establish that staleness is irrelevant.

Three representations of block identity increase RMSE by 1.2–6.0 on the detrended target, while removing spatial priors costs only 0.1. These comparisons are consistent with substantial overlap between rental-history features and geographic predictors. Small group sizes and within-group dispersion may also reduce the precision of block means. The final configuration retains the spatial priors and excludes hierarchical target encoding.

D. Macroeconomic Series

COE premiums and share prices are candidate indicators of market conditions. We aggregated COE premiums by month and category and computed monthly share-price means, log changes, and sector baskets. Adding these features increases rental RMSE by 10.0 and 10.5 in two configurations. Since the variables describe months rather than flats, we also tested them as predictors of the market index.

Ridge regressions predict the log rent index from rebased log COE premiums or equal-weight stock baskets. Each basket averages company-level log monthly adjusted prices after rebasing each company to its first month. Regressions use only pre-origin fitted indices, and forecasts are anchored to the last training-month index. Forecast-period economic covariates come from the supplied files, an availability assumption that would require forecasting those covariates in prospective use. Only the realized index used for scoring is estimated from all labeled rents.

Over the three rolling 16-month windows, mean absolute percentage error of the index level is 1.26% for the damped trend, 1.87% for the property basket, 1.90% for a random walk, and 2.61–5.14% for COE level regressions. These are index errors, not rental RMSE. A regression on 12-month log changes gives 17.30% error. Beyond a year, it recursively uses its own predicted index as the annual anchor. COE premiums and the rent index correlate at 0.91 in log levels, but both rise until 2023 and then flatten. The macroeconomic regressions do not outperform the damped forecast.

E. Handling of Categorical Attributes

The six categorical columns are town (26 levels), flat type (six), flat model (20), planning area (31), region (five), and subzone (156). The eleven-model comparison retains these source attributes. LightGBM, CatBoost, and XGBoost use library-specific categorical handling. Ridge, nearest neighbors, and the neural network use one-hot indicators. The other tree models use integer category codes, whose arbitrary order constrains threshold splits. Table 5 specifies the full preprocessing. Encoding therefore remains part of each model pipeline rather than an independently controlled factor.

Aligned category dictionaries preserve label identity between training and prediction frames, including subzones absent from an earlier training fold. No categorical level occurs only in the test file.

VII. Models and Ensemble

A. Model Families and Tuning

With identical features and splits, CatBoost [11] and LightGBM [10] lead the initial comparison of four models trained on detrended rent, at RMSEs of 521.49 and 523.32. XGBoost [12] and one-hot ridge score 528.67 and 534.26. We therefore tune the two leading libraries. This comparison does not isolate model capacity.

Random search evaluates 28 LightGBM and 22 CatBoost draws under Section III. Tree size controls interaction complexity, the learning rate scales successive corrections, and sampling and L2 penalties regularize fitting. Selected LightGBM settings are 800 trees, learning rate 0.03, 127 leaves, minimum leaf count 20, 85% row sampling, 90% column sampling, and L2 penalty 5. CatBoost uses 2,000 iterations, learning rate 0.05, depth 12, and L2 penalty 30. Frozen parameters support retraining, although the historical CatBoost winner is absent from the current sampler (Appendix K). Tuning improves LightGBM RMSE by 3.45. A later search finds no improvement on the confirmation cut.

The length of the training history also affects accuracy. Restricting training to the most recent 18 or 24 months increases the RMSE by 20.2 and 3.7, while a 36-month history and the full history perform equally. After detrending, a 2021 rental remains informative about how one flat compares with another, so we train on all labeled months.

B. Averaging Ensemble

A model with higher standalone error can improve an average when its errors offset those of other members. We evaluate weighted averaging for this reason, without fitting a stacking meta-learner.

Non-negative weight search gives 45% LightGBM and 55% CatBoost, with RMSE 518.34, versus 518.38 for equal weights. XGBoost receives no weight near the optimum. We use equal weights, averaging three LightGBM seeds and two CatBoost seeds to reduce sensitivity to initialization and sampling. Final predictions are clipped to the observed S$300–S$7,600 range. Clipping is inactive in the cached repaired, single-seed validation blend (Appendix K).

The third member is HistGB trained on unadjusted rent with block history (Section VI-B). Its rolling RMSE is 527.0, versus 519.9 for the LightGBM reference. Giving it weight 0.30 reduces rolling RMSE from 518.56 to 516.66 and confirmation RMSE from 466.54 to 463.77. The final weights are therefore 0.35 LightGBM, 0.35 CatBoost, and 0.30 HistGB, applied to predictions in dollars after index rescaling. Subsequent index and block-composition refinements give 515.73 and 462.54. These cached comparisons use one seed per library and historical inner tuning. The final refit averages seeds as above, and Appendix E reports repaired inner tuning separately. HistGB

has a different temporal specification, but its contribution to the gain was not isolated.

Further candidates, including ridge, alternative block-history features, and different weighting rules, improve the rolling folds by a few tenths but worsen the confirmation cut. Their rejection uses that cut as a secondary selection gate, not an independent evaluation.

C. Comparison of Eleven Model Classes

We compare simpler and more flexible learners to assess whether prediction after market adjustment requires sophisticated boosting models. Eleven model classes receive the same 29 source columns, comprising 23 numeric and six categorical attributes represented as in Table 5. Parameters are fixed without tuning for this comparison. Block history, hierarchical target encodings, and block composition are excluded, and a global index is used. Calendar month and approval-dependent age features retain temporal information. The explicit month counter enters only in the third treatment.

Each model is evaluated with index-adjusted rent, unadjusted rent, and unadjusted rent augmented by the explicit month counter. The unadjusted-rent treatment retains calendar month and time-dependent age features. This comparison tests whether the explicit temporal coordinate substitutes for index adjustment under otherwise fixed source attributes (Table 1).

Table 1. Each metric is averaged over three rolling folds for models trained on the same 29 source columns. Columns under With Index report index-adjusted results. The Unadj. column uses unadjusted rent, and + Month adds the month counter. Bias is the mean prediction error.

	With Index				Unadj.	+ Month
Model	**RMSE**	**MAE**	**Bias**	R^2	**RMSE**	**RMSE**
Median lookup	554.6	428.9	−38.9	0.36	965.0	n/a
Regression tree	540.6	412.6	−4.6	0.39	748.8	564.5
AdaBoost	535.9	415.1	−42.5	0.40	743.2	589.2
Ridge	534.4	410.5	−29.4	0.40	840.6	643.6
Neural network	530.8	407.8	−29.8	0.41	699.0	704.5
XGBoost	528.3	401.1	+1.7	0.42	603.7	567.2
Bagging	528.3	400.5	−1.4	0.42	658.1	555.8
Random forest	526.1	398.4	−2.1	0.42	647.2	549.7
k-nearest neighbors	523.8	397.4	−0.5	0.43	805.7	583.7
LightGBM	523.4	397.0	−0.5	0.43	611.8	556.2
CatBoost	522.3	397.8	−9.2	0.43	662.5	540.4

Under index adjustment, the best five models differ by 5.9 RMSE, and k-nearest neighbors is 1.5 above CatBoost. Several model classes therefore achieve similar accuracy under these features, encodings, and validation windows. The median lookup is 32.2 above CatBoost. Mean absolute error (MAE) changes the ranking, with LightGBM and nearest neighbors preceding CatBoost. The lookup, ridge, and neural network under-predict by S$29–S$39, whereas bagging, random forest, LightGBM, and XGBoost have mean errors within S$3. Bagging improves on the single tree by 12.4, but different leaf settings prevent attributing the gain to averaging alone.

On unadjusted rent, the range across models is 361.4 RMSE and every model under-predicts on average. Gradient boosting performs best among these configurations. Although the explicit month counter is excluded, lease age and block age retain temporal information, so this treatment does not remove all date-related predictors.

Adding the month counter does not match index-adjusted accuracy for any evaluated model. Ridge and the neural network over-predict by approximately S$290 and S$400. Tree-based models and nearest neighbors under-predict. These results describe the evaluated specifications and do not establish that other direct temporal models would perform similarly.

VIII. Error Analysis

A. Development Results

Figure 5 summarizes the historical development record. The retained sequence reduces rolling RMSE from 539.7 to 515.7 over seven stages. Because it combines feature, index, and ensemble changes, this 24-point gain is a pipeline comparison rather than a sum of independent effects. Rejected alternatives use their contemporaneous references. These comparisons show that feature and model refinements generally contribute less than index adjustment under the evaluated windows.

B. Within-Group Variation and Index Error

Rentals with identical recorded predictors can still differ in price, as illustrated in Section II. We compare this within-group variation with the selected LightGBM configuration's validation error. We also examine how error changes when validation rents help estimate the market index.

For groups containing at least two validation rows with identical predictors, estimated noise variance is the within-group sum of squares divided by the number of observations minus the number of groups (Appendix H). The denominator accounts for estimation of each group mean. Interpretation as conditional noise requires independent within-group errors around a common mean and representative group variances. These groups contain 27.6% of rolling-validation observations. Their pooled noise standard deviation is S$509, compared with model RMSE S$512 on the same rows (Table 2).

An oracle variant estimates the index using training and validation rents before refitting the model. The Noise column is estimated within-group variance divided by model mean squared error (MSE), expressed as a percentage. Oracle is the MSE reduction obtained with the oracle index, divided by model MSE. Resid. equals 100% minus Noise minus Oracle. All three shares use repeated-group rows and sum to 100% before rounding. Their pooled values are 98.8%, 1.1%, and 0.1%, respectively. They do not identify independent causal error sources. Negative residual shares reflect estimation uncertainty rather than negative error variance.

The estimated noise level is close to this configuration's error on repeated-group rows. Different attributes, learners, or temporal assumptions could still improve prediction. Similar aggregate errors on grouped and ungrouped rows do not establish equal conditional noise, and these estimates do not establish a floor for other rows or the hidden test set.

The oracle reduces all-row RMSE by 1.5–4.2 across rolling folds, compared with the 86.4-point cost of omitting index adjustment. This suggests that rescaling historical rents provides

Table 2. Error decomposition is reported for the selected LightGBM configuration. Under All Rows, Oracle is RMSE using an index fitted on training and validation rents. Under Rows in Repeated Groups, Noise SD is the estimated within-group standard deviation, and Noise, Oracle, and Resid. are the MSE shares defined in Section VIII. RMSE and Noise SD are in Singapore dollars. The summary row averages fold RMSEs for All Rows and pools forecast instances for repeated groups.

	All Rows		Rows in Repeated Groups				
Fold	**RMSE**	**Oracle**	**RMSE**	**Noise SD**	**Noise**	**Oracle**	**Resid.**
1	554.7	553.1	544.9	542.5	99.1%	0.3%	0.5%
2	519.2	514.9	511.5	506.1	97.9%	2.2%	0.0%
3	482.6	481.1	476.3	475.4	99.6%	0.8%	−0.4%
Summary	519.9	517.5	512.0	509.0	98.8%	1.1%	0.1%
Confirm	465.5	463.1	456.5	457.4	100.4%	0.4%	−0.8%

most of the measured benefit under these windows. Because the oracle uses validation targets and changes the transformed model, its improvement is an optimistic diagnostic rather than a formal bound on forecasting gains.

Estimated within-group noise standard deviations are S$279, 471, 588, 479, and 443 for 2021 through early 2025. Conditional dispersion therefore varies over time. This may contribute to differences between validation periods, but it does not explain the lower public score because test-period noise is unobserved.

C. Error by Flat Type and Limitations

RMSE increases from S$483 for 3-room to S$624 for executive flats, while relative RMSE decreases from 22% of mean rent for 2-room to 16% for 5-room flats. Mean errors range from S$34 over-prediction for 2-room flats to S$20 under-prediction for executive flats. The 1-room group has only 46 fold observations and mean error S$115 (Table 8). Because evaluation uses squared errors in dollars, the same proportional error receives a larger penalty when rent is higher.

Unobserved flat attributes can explain variation classified here as noise, limiting the interpretation of repeated-group estimates. All rolling origins follow the 2022 surge, so performance during renewed rapid growth remains untested. The confirmation cut influenced selection, and overlapping validation windows do not independently confirm small improvements. The historical block-history member also used row-based inner splits and history features constructed before those splits. The repaired protocol uses whole-month origins and rebuilds history within each inner fold. It changes member rolling RMSE from 527.0 to 526.4 and blend RMSE from 516.66 to 516.54 (Appendix E).

D. Implications for Stakeholders

Potential renters can use predictions to compare asking rents conditional on recorded attributes. The estimated S$509 noise standard deviation describes variation within repeated feature groups, not a calibrated prediction interval for an individual listing. Flat condition and tenancy terms remain unobserved.

Landlords and real estate agents can use these estimates as benchmarks for asking rents. Town and flat type have greater held-out permutation importance than nearest-MRT distance, but this measures predictive reliance rather than causal effects or returns to advertising. The indices also distinguish temporal movements across flat types, rising by 46% for 2-room and 61% for executive flats over the labeled period.

The Singaporean government can use composition-adjusted rent trends to inform housing supply and rental assistance policies. The estimated index adjusts for observed block and flat-type composition under the stable-base assumption. Neither this adjustment nor the measured changes in within-group dispersion identify the effects of a cooling measure.

E. Future Work

Additional chronological origins should include renewed rent growth to test damping, with a later evaluation window reserved before model selection. Small differences between leading models should be assessed by paired resampling of contiguous calendar-month blocks, retaining all forecasts of an approval together across overlapping windows. Floor level, flat condition, tenancy duration, and included services should be collected where available at prediction time. Re-estimating within-group variation after adding these attributes would test whether omitted information explains part of the estimated noise. Additive and multiplicative decompositions should receive separate hyperparameter searches under equal validation budgets. Geographic features should incorporate station opening dates to represent the operating network in each approval month. Prediction intervals should also be evaluated for coverage by forecast horizon and flat type before use for individual rental decisions.

IX. Conclusion

Under the evaluated forecast windows, index adjustment contributes more to accuracy than tuning LightGBM or choosing among the leading model classes. Removing the index increases rolling RMSE by 86.4 in a matched LightGBM comparison, whereas tuning improves it by 3.5. The five leading classes differ by 5.9 under a fixed index setup. Nearest-neighbor regression achieves accuracy close to the leading boosting models after market adjustment, indicating the importance of temporal treatment for this dataset.

On the 27.6% of validation rows in repeated feature groups, estimated pooled noise variance equals 98.8% of one LightGBM configuration's MSE. This result depends on the grouping and estimation assumptions and does not establish a test-set error floor. The ensemble scores S$431.01 at index shrinkage 0.5, and its 0.7 variant achieves our best public RMSE of S$430.76.

AI Declaration

Claude (Anthropic) was used for brainstorming, checking ideas, explaining lecture concepts, as a coding assistant, to clean code, and as a writing assistant and for version management. We are responsible for the content and quality of the submitted work.

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APPENDIX

A. Workload Breakdown

Table 3 lists the contribution of each member.

Table 3. The workload is summarized by team member and primary responsibility.

Member	Main Responsibilities
Angeles Mathena	Modeling pipeline, market index and its forecast, ablation study, error decomposition, and final ensemble
Alzamil Nojood Ahmad	Feature extensions, macroeconomic series, and index variants
Su Yubo	Data integration, geographic join repair, block history features, validation design, and the model comparison
Szentpeteri David	Exploratory analysis, data cleaning, first feature set, and baseline models
Dai Peisong	Exploratory analysis, feature construction, model comparison, and ablation study

B. Validation Splits

Table 4. The first three rows describe the rolling validation windows, and the last row describes the confirmation window. Lengths are given in months.

Training Months	Validation Months	Length	Rows
Jan 2021 to Mar 2023	Apr 2023 to Jul 2024	16	48,301
Jan 2021 to Jul 2023	Aug 2023 to Nov 2024	16	47,036
Jan 2021 to Nov 2023	Dec 2023 to Mar 2025	16	47,171
Jan 2021 to Mar 2024	Apr 2024 to Mar 2025	12	35,158

C. Representation of the Categorical Attributes

Table 5. The six categorical columns are represented separately for each model family in the eleven-model comparison.

Model Family	Representation
LightGBM, CatBoost, XGBoost	Library-specific categorical handling, without treating category labels as numeric magnitudes.
Regression tree, random forest, bagging, AdaBoost	Ordinal integer codes. Unseen levels map to -1. The arbitrary ordering constrains threshold splits and may require extra splits to represent category-specific effects.
Ridge, nearest neighbors, neural network	One-hot indicators, levels appearing fewer than 30 times pooled into one column, with median imputation and standardization of the numeric columns. These three would read an integer code as a magnitude.

D. Permutation Importance of the Gradient-Boosting Baseline

Table 6. Importance is the decrease in held-out R^2 after one feature is shuffled in the chronological baseline holdout.

Feature	Importance
Town	0.352
Flat type	0.261
Floor area	0.134
Lease age	0.061
Flat model	0.021
Distance to nearest MRT station	0.012
Distance to nearest mall	0.007
Distance to nearest primary school	0.001
Calendar month	0.0003
Month counter	0 (exact)
Mean stock closing price	-0.00001
Mean COE premium	-0.00003

E. Block Rent History by Fold

Table 7. RMSE and mean prediction error are reported for gradient boosting trained on unadjusted rent with and without block history. The final row reports the confirmation cut.

Validation Months	RMSE		Mean Error	
	Without	With	Without	With
Apr 2023 to Jul 2024	585.4	572.8	−139.8	−120.8
Aug 2023 to Nov 2024	532.5	521.7	−36.1	−8.7
Dec 2023 to Mar 2025	499.5	486.6	−64.6	−16.4
Mean of the three	539.1	527.0	−80.1	−48.7
Apr 2024 to Mar 2025	481.5	467.8	−56.5	−28.8

The **With** column and the submitted blend use positional `TimeSeriesSplit` for inner tuning. Its boundaries can divide a month, evaluating within-month interpolation as well as future prediction. Block-history features were also constructed before inner splitting and could include inner-validation rents. The repaired protocol uses whole-month origins and rebuilds history from each inner training frame. Its horizon is 16 months where feasible and 14 at the earliest cut. Both protocols use identical sampled candidates. The comparison measures the combined repair of inner splits and history construction.

The repair changes the selected setting at two of the four origins, cuts 27 and 39, where the positional search had selected a learning rate of 0.03 rather than 0.1. At the other two origins it selects the same setting and the fold predictions are identical. Where the selection changes, the member improves by 1.87 at cut 27 and 0.22 at cut 39, so its rolling mean moves from 527.02 to 526.40 and the confirmation cut from 467.76 to 467.54. The ensemble moves from 516.66 to 516.54 on the folds and from 463.77 to 463.82 on the confirmation cut, and 30% remains the best weight.

In this comparison, the repair improves the member's outer-validation scores where selection changes. The ensemble changes by less than 0.2 RMSE on either protocol. This sensitivity analysis does not establish that inner leakage generally has negligible effects. `src/h2h.py --inner rows` reproduces the historical column. `--inner month` writes the repaired results to `outputs/ablation_head_to_head_repaired.csv`.

F. Error by Flat Type

Table 8. Errors for the selected LightGBM configuration are pooled over the rolling folds and reported by flat type. Overlapping validation windows can count an approval more than once.

Flat Type	Rows	Mean Rent	RMSE	Bias	RMSE / Rent
1-room	46	1,596	439.8	+114.6	27.6%
2-room	3,637	2,184	485.0	+34.3	22.2%
3-room	46,708	2,696	483.3	+18.9	17.9%
4-room	51,844	3,237	527.4	−12.9	16.3%
5-room	32,901	3,440	536.3	−17.3	15.6%
Executive	7,372	3,577	623.7	−19.6	17.4%

G. Mean Test-Period Index Under Each Forecast

Table 9 gives the values behind the 1.5% range in Section V-B, computed by `src/index_horizon.py`. The mean index is taken over the 50,000 test rows, so it is weighted by the number of rows at each horizon. The shift is the difference between each mean index and that of the submitted forecast, multiplied by the mean training rent of S$2,719.

Table 9. Each mean index is weighted by the number of test rows at each forecast horizon. The reported shift is the difference from the submitted forecast's mean index, multiplied by the mean training rent of S$2,719.

Forecast	Mean Index	Shift (S$)
Random walk	1.0000	−6.6
Damped, 12 months, ϕ = 0.4	1.0011	−3.5
Damped, 12 months, ϕ = 0.6 (shipped)	1.0024	0.0
Damped, 12 months, ϕ = 0.75	1.0043	+5.2
Damped, 12 months, ϕ = 0.85	1.0068	+11.8
Damped, 12 months, ϕ = 0.95	1.0113	+24.0
Damped, 18 months, ϕ = 0.85	1.0063	+10.6
Linear, 12 months	1.0150	+34.2
Linear, 18 months	1.0141	+31.7

H. Pooled Variance Estimator

For groups g containing $n_g \geq 2$ rows with identical predictors, the pooled variance estimator is

$$\widehat{\sigma}^2 = \frac{\sum_g \sum_{i \in g} (y_i - \bar{y}_g)^2}{\sum_g (n_g - 1)}.$$

The denominator removes one degree of freedom per fitted group mean. Groups are defined separately within each validation fold to avoid counting overlapping observations as additional within-group matches. This estimator describes pooled within-group variation. Interpreting it as the noise level for other observations requires additional representativeness assumptions.

Multiplying the training variance estimate by the proportion of rows in repeated groups and taking the square root gives approximately S$240. Applying it to the test-group proportion gives approximately S$245. These illustrative scales assume zero error on all remaining rows. The test calculation additionally transfers historical noise into an unobserved period. Neither calculation is an empirical test-set RMSE or a rigorous lower bound.

I. The Distribution of Rent

Figure 3 uses all labeled rentals to match the training sample analysed in Section II-B.

J. Detailed Ablation Results

Figure 5 summarizes the ablation record of Section VIII-A, showing the retained stages and the cost of the rejected alternatives. Figure 4 gives the results for individual candidates. Panel (a) is the feature-group ablation, panel (b) gives the gradient-boosting configurations by fold, and panel (c) compares the rolling-fold score of each candidate with its confirmation-cut score.

K. Reproducing Each Result

The repository contains scripts, saved results, and reproduction instructions in `README.md`. First, `src/data.py` builds `outputs/base.parquet`. Analyses then read this feature

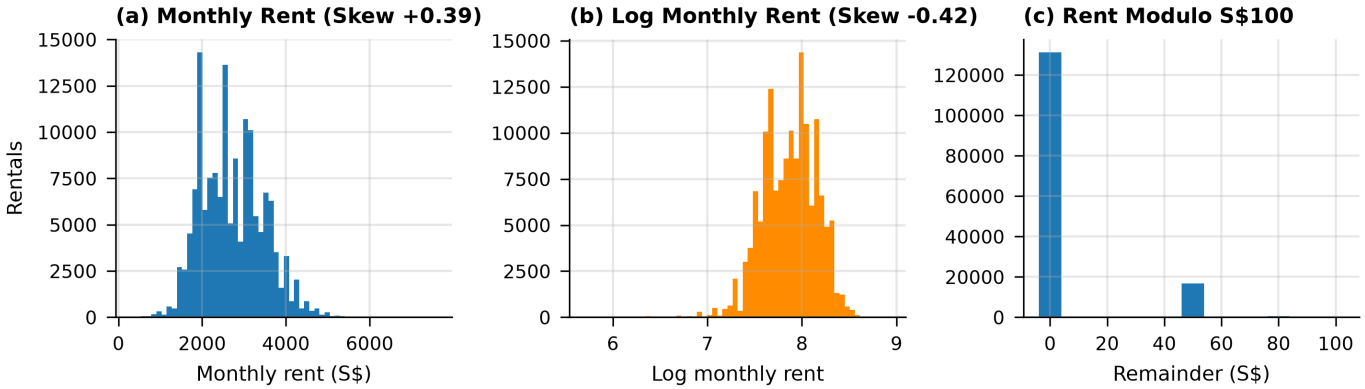


Figure 3. All panels use the 150,000 labeled rentals. Panel (a) shows the mildly right-skewed rent distribution. Panel (b) shows the left-skewed distribution after logarithmic transformation. Panel (c) shows that rents follow a coarse grid, with 98.7% being exact multiples of S$50.

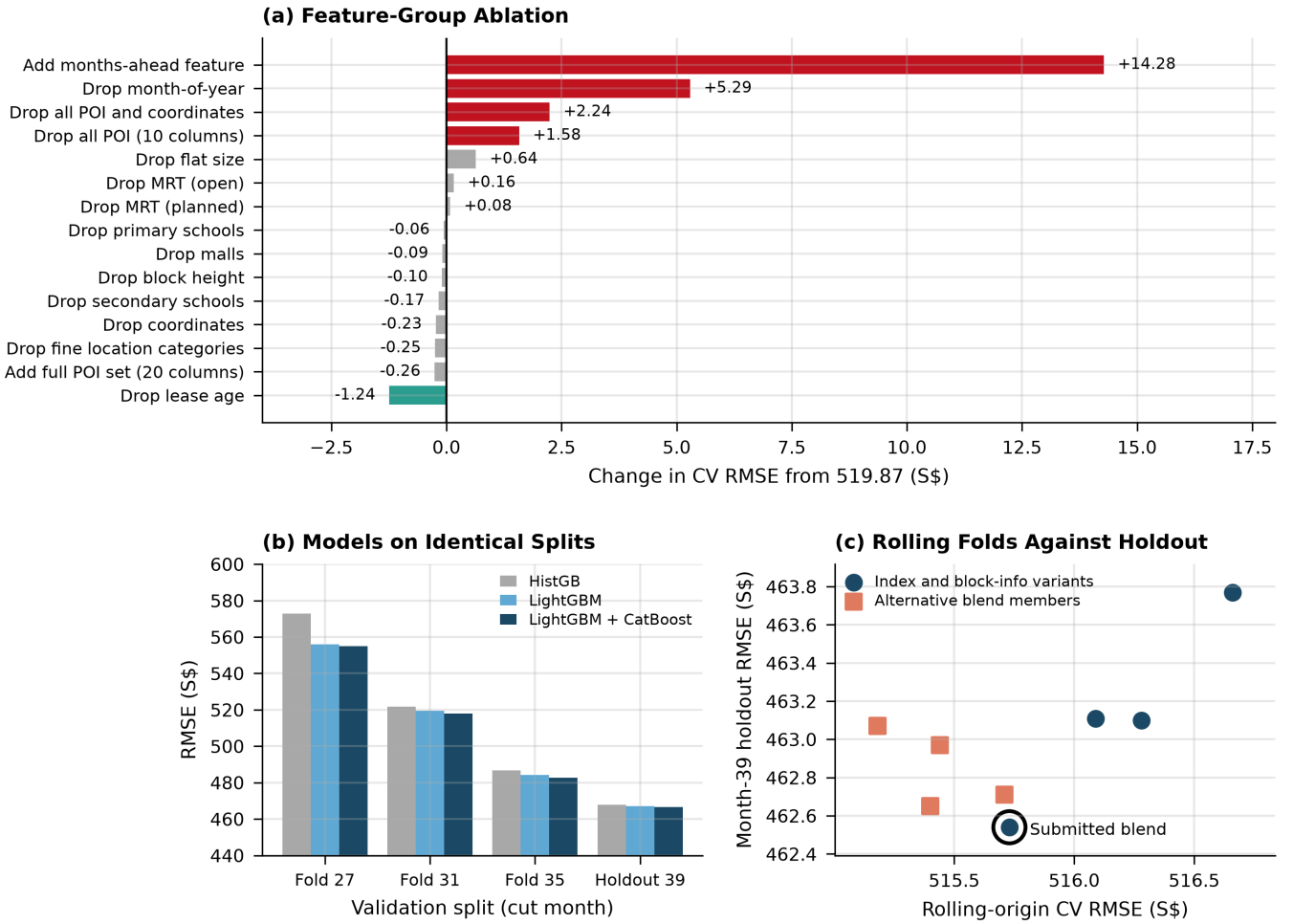


Figure 4. Panel (a) shows the RMSE change after adding or removing each feature group relative to the tuned LightGBM reference of 519.87. Panel (b) compares three configurations across validation splits, including the March 2024 confirmation cut. Panel (c) compares candidates' rolling-fold and confirmation-cut RMSEs.

table. Ensemble analyses have additional dependencies. Scripts `h2h.py`, `confirm.py`, and `members.py` generate caches used by the blend scripts. Figure scripts run after their input CSV files exist. Table 10 maps results to scripts and output files. Historical replay limitations are specified below.

For historical ablation tables, `src/groups.py` maps each named comparison to registry configurations. `run_ablation.py --group` *name* re-evaluates the group un-

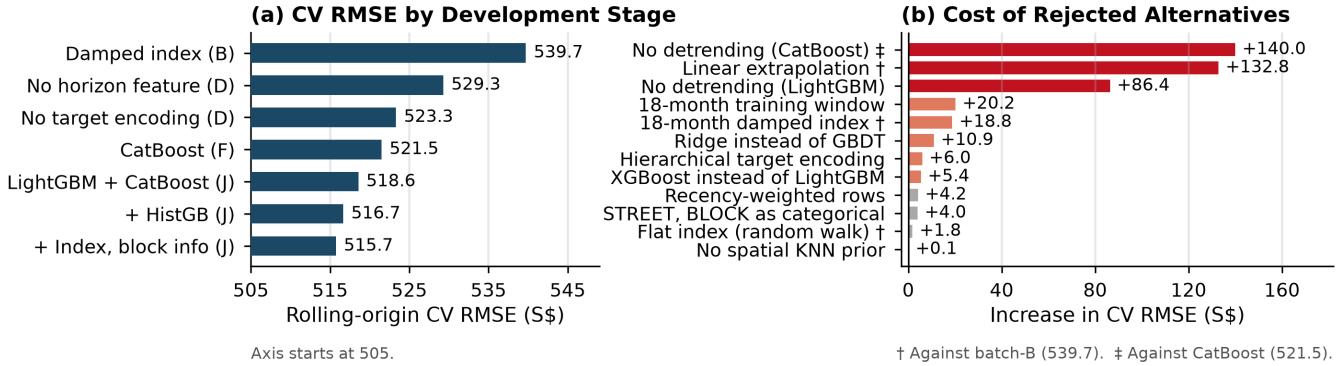


Figure 5. Panel (a) shows rolling-origin RMSE after each retained development stage, labeled by ablation batch. Panel (b) shows the RMSE increases for rejected alternatives. The reference is 523.3 except for batch B at 539.7 (†) and CatBoost at 521.5 (‡).

Table 10. Each reported result is mapped to its generating script and saved output.

Where	Script in `src/`	Output in `outputs/`
Sec. II, App. I	`eda.py`	`eda_summary`
Sec. IV, App. D	`baseline.py`	`permutation_importance`
Sec. V-B, Figure 2	`run_ablation.py`	`ablation_index_forecast, ablation_index_window_damping`
Sec. V	`additive_check.py`	`additive_index_check`
Sec. V-B, App. G	`index_horizon.py`	`index_test_horizon`
Sec. V-C	`run_ablation.py, index_horizon.py`	`ablation_index_segments, index_by_flat_type`
Sec. VI	`run_ablation.py, altreps.py`	`ablation_feature_groups, ablation_location_representation, ablation_altreps`
Sec. VI-B, App. E	`blockhist.py, h2h.py`	`ablation_blockhist, ablation_head_to_head`
Sec. VI-C	`run_ablation.py`	`ablation_target_encoding, ablation_lagged_block_history`
Sec. VI-D	`macro.py`	`macro_backtest`
Sec. VII	`run_ablation.py, tune.py`	`ablation_model_family, tuning, ablation_retune`
Sec. VII-B	`blend_weights.py, blend_members.py`	`ablation_blend_weights, ablation_blend_members`
Sec. VII-C, Table 1	`ladder.py`	`ladder_metrics, ladder_by_horizon`
Sec. VIII, Table 2	`confirm.py, error_decomp.py`	`ablation_confirmation_cut, error_budget`
Sec. VIII, Apps. F, H	`residuals.py`	`residual_slices, residual_by_horizon, rmse_floor, feature_floor`
Sec. V	`compare_submissions.py`	`submission_comparison`
Final predictions	`predict.py`	`submissions/`

der current defaults and writes a separate `_rerun` file. Batch B preserves the ranking but changes the scores. Its flat-index configuration gives 526.6 instead of 541.5. Batch D reproduces all nine rows at their recorded precision, including fold scores and bias, after explicitly specifying the historical encoding mode. `--list-groups` reports coverage. Seven groups contain registered configurations only, while four also contain historical variants absent from the registry. Encoder settings must be explicit because smoothing and recency parameters are inactive when target encoding is disabled. The original tables retain the comparisons used during development.

Forty of the fifty historical hyper-parameter draws regenerate from the current sampler and seed. Ten draws, including the selected CatBoost configuration, predate that sampler. Frozen parameters in `src/tuned_params.py` support retraining the reported models.

Across overlapping validation windows, clipping altered none of the 177,666 cached forecast instances from the repaired, single-seed validation blend. Predictions range from S$1,278 to S$5,659, within the S$300–S$7,600 clipping bounds, so clipping leaves this cached blend's RMSE unchanged.

The separate 71-feature ridge pipeline scores 505.60 publicly. Its predictions correlate with our ensemble at 0.973 within test months, but its monthly mean prediction rises by 0.80% per month rather than 0.02% and ends 15% higher. Both the model and temporal specification differ, so these differences do not identify the effect of index forecasting.

Repeated MRT coordinates leave nearest distance unchanged but affect second-nearest distance and radius counts on 21% and 27% of rows, respectively. Deduplication and removal of the planned Terminal 5 station is a sensitivity check, not a reconstruction of the operating network.